

# The Local Convergence Rate of Gaussian Natural Gradient Flow is Empirically Dimension-Free

## Abstract

The Gaussian natural gradient flow contracts toward its equilibrium at a local rate  $\gamma_{\text{loc}}$  equal to the smallest eigenvalue of the linearized positive generator  $L_\star$  in the Fisher–Rao metric. The current analysis bounds  $\gamma_{\text{loc}}$  from below by  $1/(4+\Gamma)$  with  $\Gamma = \min\{\beta - 1, n(3 + \frac{4}{\sqrt{\pi}}(1 + \log \kappa))\}$ , whose second branch carries an explicit dimension factor  $n = N_\theta$ . We estimate  $\gamma_{\text{loc}}$  directly from the linearized operator over a controlled grid ( $N_\theta = 1, \dots, 16$ ;  $\kappa \in \{2, 5, 10, 20, 50, 100\}$ ; five potential families; three seeds;  $M = 2^{22}$  Monte Carlo samples per point, float64, GPU). Across every family the measured  $\gamma_{\text{loc}}$  is essentially flat in  $N_\theta$  and varies only with the conditioning  $\kappa$ . This indicates that the genuine local rate is governed by  $\kappa$  rather than  $N_\theta$ , and that the  $N_\theta$  factor in the present bound is an artifact of the proof rather than a property of the flow. We are careful to separate this conclusion, which rests on the well-conditioned estimator  $\gamma_{\text{loc}}$ , from the full-operator norm estimate  $\hat{\Lambda}$ , whose apparent growth with  $N_\theta$  we show to be finite-sample spectral noise.

## 1 Setup

Work in equilibrium-whitened coordinates, where the target satisfies  $\mathbb{E}_{\mathcal{N}(0,I)}[\nabla V(Z)] = 0$  and  $\mathbb{E}_{\mathcal{N}(0,I)}[\nabla^2 V(Z)] = I$ , so the equilibrium of the Gaussian natural gradient flow is  $a_\star = (0, I)$ . Parameterize a perturbation as  $m = u$  and  $C = I + X$  with  $X \in \text{Sym}(N_\theta)$ .

**Definition 1** (Linearized operators). For  $u \in \mathbb{R}^{N_\theta}$  and  $X \in \text{Sym}(N_\theta)$ , with  $Z \sim \mathcal{N}(0, I)$  and  $W(Z) = \nabla^2 V(Z)$ , define

$$\begin{aligned} T[X] &= \mathbb{E}[\text{Tr}(X W(Z)) Z], & T^*[u] &= \mathbb{E}[W(Z) (Z^\top u)], \\ H[X] &= \mathbb{E}[W(Z) (Z^\top X Z - \text{Tr } X)], & H_{\text{sym}}[X] &= \tfrac{1}{2}(H[X] + H^*[X]). \end{aligned}$$

**Proposition 1** (Linearized generator). *The linearized positive generator  $L_\star = -J_\star$  of the flow is*

$$L_\star(u, X) = \left( u + \tfrac{1}{2}T[X], \quad X + T^*[u] + \tfrac{1}{2}H_{\text{sym}}[X] \right), \quad (1)$$

*and it is self-adjoint with respect to the local Fisher–Rao inner product at equilibrium,*

$$\langle (u_1, X_1), (u_2, X_2) \rangle_\star = u_1^\top u_2 + \tfrac{1}{2} \text{Tr}(X_1 X_2).$$

*The local convergence rate is the smallest eigenvalue  $\gamma_{\text{loc}} = \lambda_{\min}(L_\star)$ .*

The analysis of the manuscript establishes the lower bound

$$\gamma_{\text{loc}} \geq \frac{1}{4 + \Gamma}, \quad \Gamma = \min \left\{ \beta - 1, \quad N_\theta \left( 3 + \frac{4}{\sqrt{\pi}}(1 + \log \kappa) \right) \right\}, \quad (2)$$

where  $\alpha \leq 1 \leq \beta$  are the strong-convexity and smoothness constants and  $\kappa = \beta/\alpha$ . The dimension-free conjecture replaces  $\Gamma$  by  $D_\kappa = (4 + \frac{4}{\sqrt{\pi}})(1 + \log \kappa)$ , i.e. a rate that depends on the conditioning  $\kappa$  but not on  $N_\theta$ . The experiments below test which behavior the measured  $\gamma_{\text{loc}}$  actually follows.

## 2 Estimator and experimental design

**Estimator.** The operators of Definition 1 are estimated on a fixed Monte Carlo bank  $\{Z_j\}_{j=1}^M$ ,  $Z_j \sim \mathcal{N}(0, I)$ . Two corrections are essential for the spectrum to be meaningful on a finite bank.

- *Symmetrization.* In exact expectation  $H$  is self-adjoint, but the plain forward Monte Carlo estimator is not; its Frobenius adjoint is a different operator. We estimate the explicitly symmetrized  $H_{\text{sym}} = \frac{1}{2}(H_{\text{fwd}} + H_{\text{adj}})$  on the same bank and use it for every eigenvalue. The residual self-adjointness error is at the level of  $10^{-15}$ .
- *Fisher–Rao packing.*  $L_\star$  is self-adjoint in the Fisher–Rao metric, not the Euclidean one. Packing tangents as  $y = (u, \text{sym\_to\_vec}(X)/\sqrt{2})$ , with  $\text{sym\_to\_vec}$  a Frobenius isometry, makes the packed matrix genuinely symmetric, and  $\gamma_{\text{loc}}$  is obtained from a dense symmetric eigensolve.

The revised proof route also records the first-Hermite coupling

$$\tau_H^2 = \|T\|_{\text{op}}^2 = \sup_{\|w\|=1} \left\| \mathbb{E}[\nabla^2 V(Z)(w^\top Z)] \right\|_F^2.$$

This is the quantity that controls the mean-covariance coupling in the linearized generator. The deterministic reduction  $\gamma_{\text{loc}} \geq 1/(\tau_H^2 + 3)$  separates this coupling channel from large positive covariance-curvature modes, which may increase  $\hat{\Lambda}$  without slowing the flow. The same Bochner-Schur identity and the upper Hessian bound give the baseline  $\tau_H^2 \leq \beta$ , hence  $\gamma_{\text{loc}} \geq 1/(\beta + 3) \geq 1/(\kappa + 3)$  in whitened coordinates. The remaining question is whether  $\beta$  can be replaced by  $O(1 + \log \kappa)$ . The implementation also decomposes the top singular mode of  $T$  into the longitudinal, mixed, and transverse blocks of Section 3, so future adversarial runs can identify whether large coupling is entering through the only unresolved high-rank transverse channel. For separable control potentials a Gauss–Hermite quadrature provides an exact, sample-free benchmark  $\Lambda_{\text{exact}} = \max_i \mathbb{E}[V_i''(Z)(Z^2 - 1)]$  that is dimension-free by construction; it is used below to expose finite-sample noise in the full-operator estimate.

**Potential families.** All non-Gaussian targets are built by the same construction: a raw feature  $\Phi$  is centered so that  $\mathbb{E}_{\mathcal{N}(0, I)}[\nabla V] = 0$  and  $\mathbb{E}_{\mathcal{N}(0, I)}[\nabla^2 V] = I$ , giving  $V = \frac{1}{2}\|\theta\|^2 + \rho(\Phi - b^\top \theta - \frac{1}{2}\theta^\top M \theta)$  with  $\nabla^2 V = I + \rho(\nabla^2 \Phi - M)$ . The coupling  $\rho$  is set from the target conditioning  $\kappa$  through  $\rho = \frac{\kappa-1}{\kappa+1}/L_A$ , so that  $\alpha = \frac{2}{\kappa+1}$ ,  $\beta = \frac{2\kappa}{\kappa+1}$ , and  $\beta/\alpha = \kappa$ . The five families are chosen to vary one structural property of  $\nabla^2 V$  at a time, so that any genuine dimension dependence would have to survive across qualitatively different Hessian geometries:

- **Gaussian** ( $V = \frac{1}{2}\|\theta\|^2$ ,  $\nabla^2 V = I$ ). The exact control: the equilibrium is reached identically, the true operators vanish, and  $\gamma_{\text{loc}} = 1$  analytically. It calibrates the estimator and the finite-sample floor, since any departure from  $\Lambda = 0$ ,  $\gamma_{\text{loc}} = 1$  is pure Monte Carlo noise.
- **Separable**,  $\Phi(\theta) = \sum_i \phi(\theta_i + c_i)$  with  $\phi = \log \cosh$ . The Hessian is *diagonal*, so the linearized operator decouples coordinate-wise and the rate is governed by the one-dimensional curvature law  $\mathbb{E}[V_i''(Z)(Z^2 - 1)]$ . This is the only family with a closed-form, dimension-free benchmark (Gauss–Hermite), making it the reference case: if the rate were to acquire an  $N_\theta$  factor, the diagonal family would expose it most cleanly. It is also the most strongly conditioned in practice (smallest realized  $\gamma_{\text{loc}}$ ).

**Table 1:** Metadata of the final natural-gradient local-rate production run. This is the only run interpreted as evidence; exploratory smoke/baseline/pilot runs are excluded.

| quantity                 | value                                                            |
|--------------------------|------------------------------------------------------------------|
| dimensions $N_\theta$    | $1, \dots, 16$ (16 values)                                       |
| conditioning $\kappa$    | $\{2, 5, 10, 20, 50, 100\}$                                      |
| families                 | Gaussian, separable, additive-index, random-feature, radial-tail |
| seeds                    | $\{0, 1, 2\}$                                                    |
| rows                     | 1440 (all status ok)                                             |
| Monte Carlo samples $M$  | $4194304 = 2^{22}$                                               |
| estimator                | symmetrized ( $H_{\text{sym}}$ )                                 |
| backend / device / dtype | torch / NVIDIA H200 NVL / float64                                |
| run id                   | nglr_20260605_224942                                             |

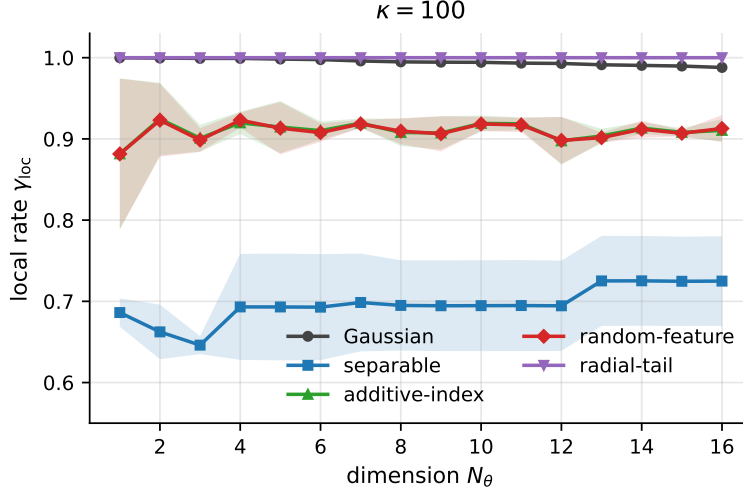
- **Additive-index**,  $\Phi(\theta) = \frac{1}{\sqrt{r}} \sum_{\ell=1}^r a_\ell \phi(w_\ell^\top \theta + c_\ell)$  with  $r = 4N_\theta$  random unit directions  $w_\ell$ . The Hessian  $\nabla^2 \Phi = \sum_\ell (\text{scalar}) w_\ell w_\ell^\top$  is a sum of rank-one terms along fixed directions; it carries genuine off-diagonal curvature, so the linearized operator no longer decouples. This is the first test of whether coupling between coordinates introduces dimension dependence.
- **Random-feature**: the same ridge-sum form with a dense bank of random directions and width growing as  $r = 4N_\theta$ . It is the general coupled potential and the primary stress test—the Hessian is full and its spectrum is determined by the random-feature geometry rather than a small set of indices.
- **Radial-tail**,  $\Phi(\theta) = h(\|\theta\|^2)$  with  $h = \log \cosh$ . The Hessian  $2h'I + 4h''\theta\theta^\top$  is isotropic in distribution, so the perturbation lives almost entirely in the trace/volume mode. It probes the heavy-tailed, rotationally symmetric regime; because the isotropic mode is exactly the channel  $L_\star$  contracts fastest, it serves as a second (non-separable) near-ideal control.

These families span the relevant axes—no coupling (Gaussian, separable), sparse directional coupling (additive-index), dense coupling (random-feature), and isotropic heavy-tail (radial-tail)—so that a dimension-dependent rate would have to appear in at least one of them if it were real.

**Final production grid.** All results in this report come from a single production run; smoke, baseline, and high-dimensional pilot runs are explicitly excluded. The grid spans five potential families (the analytic Gaussian; separable; additive-index; random-feature; radial-tail),  $N_\theta = 1, \dots, 16$ , six conditioning values  $\kappa$ , and three seeds, with  $M = 2^{22}$  Monte Carlo samples per grid point in float64 on a single GPU. Every one of the 1440 rows completed successfully. Metadata is summarized in Table 1.

### 3 Results

$\gamma_{\text{loc}}$  **is flat in  $N_\theta$ .** Figure 1 shows  $\gamma_{\text{loc}}$  as a function of  $N_\theta$  at the hardest conditioning  $\kappa = 100$ , for each family (mean over seeds, shaded  $\pm$  standard deviation). For every family the curve is essentially horizontal across  $N_\theta = 1, \dots, 16$ : the Gaussian and radial-tail controls sit at  $\gamma_{\text{loc}} \approx 1$ , the coupled additive-index and random-feature families hold near 0.91, and the separable family near 0.73, with no downward drift in  $N_\theta$ . This is the central observation: the measured rate does



**Figure 1:** Local rate  $\gamma_{loc}$  versus dimension  $N_{\theta}$  at  $\kappa = 100$ , by potential family (mean over three seeds; shaded band  $\pm 1$  s.d.). Every family is flat in  $N_{\theta}$ , contradicting a genuine  $1/N_{\theta}$ -type dimension dependence of the rate.

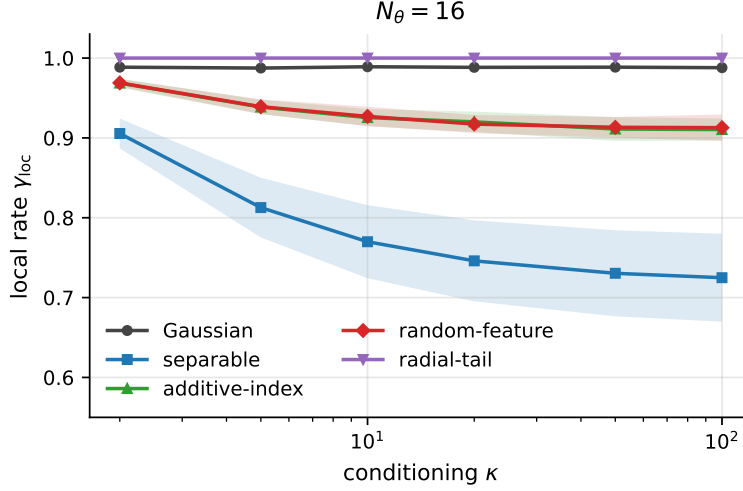
not decay with dimension, in contrast to the  $1/(4 + \Gamma)$  branch of (2), whose  $N_{\theta}$  factor would force  $\gamma_{loc} \rightarrow 0$  as  $N_{\theta}$  grows.

$\gamma_{loc}$  **varies with  $\kappa$ .** Figure 2 fixes  $N_{\theta} = 16$  and varies  $\kappa$ . Here the coupled families do move: the separable rate decreases from  $\approx 0.91$  at  $\kappa = 2$  to  $\approx 0.73$  at  $\kappa = 100$ , and the additive-index/random-feature families from  $\approx 0.97$  to  $\approx 0.91$ , while the Gaussian and radial-tail controls remain at  $\approx 1$ . The dependence is mild and monotone in  $\log \kappa$ , consistent with the conditioning-driven  $D_{\kappa}$  form rather than a dimensional one. Table 2 gives the per-family ranges over the whole grid.

**Per-family reading.** The ordering of the families by  $\gamma_{loc}$  follows directly from their Hessian geometry, which is the consistency check that the numbers are measuring the intended effect.

- The *Gaussian* and *radial-tail* families sit at  $\gamma_{loc} \approx 1$  throughout. For the Gaussian this is exact; for radial-tail the perturbation is isotropic and therefore lives in the trace/volume mode, which is precisely the channel  $L_{\star}$  contracts at unit rate. Their flatness in both  $N_{\theta}$  and  $\kappa$  confirms the estimator is unbiased where the answer is known.
- The *separable* family has the smallest rate (down to  $\gamma_{loc} \approx 0.65$ ) and the strongest  $\kappa$ -dependence. This is expected: it is the most strongly conditioned realization (at  $\kappa = 100$ ,  $N_{\theta} = 8$  the centering gives  $\rho \approx 1.55$ ,  $\alpha \approx 0.02$ ,  $\beta \approx 1.98$ ), and its diagonal Hessian concentrates the curvature deficit in the shape modes that  $L_{\star}$  contracts slowest. Even so  $\gamma_{loc}$  is flat in  $N_{\theta}$ .
- The *additive-index* and *random-feature* families track each other closely ( $\gamma_{loc} \approx 0.91$  at  $\kappa = 100$ ) and lie between the controls and the separable case. The off-diagonal curvature they introduce lowers the rate relative to the isotropic controls but—crucially—adds no  $N_{\theta}$  dependence, even as the feature width grows with dimension ( $r = 4N_{\theta}$ ).

In every family the dependence is on  $\kappa$ , through how aggressively the target is conditioned, and not on  $N_{\theta}$ .



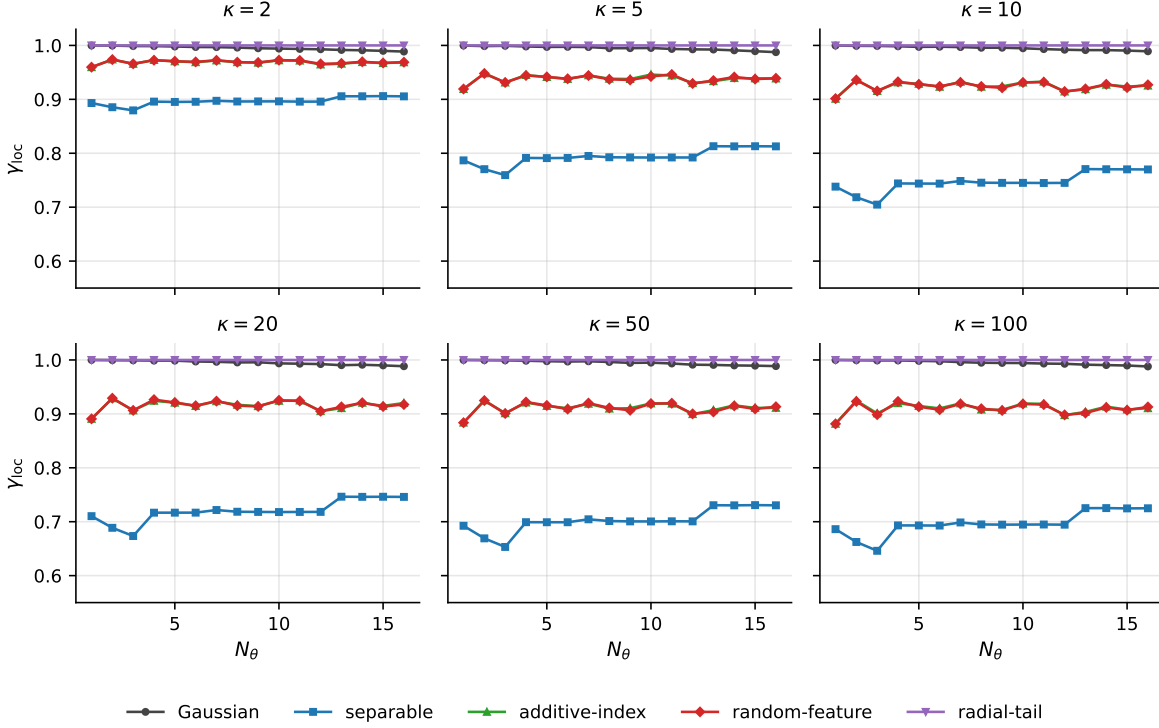
**Figure 2:** Local rate  $\gamma_{\text{loc}}$  versus conditioning  $\kappa$  at  $N_\theta = 16$  (log-scaled  $\kappa$ ). The rate decreases mildly with  $\kappa$  for the coupled families and is flat for the controls—the opposite of the  $N_\theta$  behavior in Figure 1.

**Table 2:** Range of the seed-averaged  $\gamma_{\text{loc}}$  over the full grid, by family, and the largest inverse rate  $1/\gamma_{\text{loc}}$ . The rate stays well away from 0 in every family; the spread within a family is driven by  $\kappa$ , not  $N_\theta$ .

| family         | min $\gamma_{\text{loc}}$ | max $\gamma_{\text{loc}}$ | max( $1/\gamma_{\text{loc}}$ ) |
|----------------|---------------------------|---------------------------|--------------------------------|
| Gaussian       | 0.987                     | 1.000                     | 1.013                          |
| separable      | 0.646                     | 0.906                     | 1.548                          |
| additive-index | 0.882                     | 0.974                     | 1.134                          |
| random-feature | 0.881                     | 0.974                     | 1.134                          |
| radial-tail    | 1.000                     | 1.000                     | 1.000                          |

**Relation to the bound.** Two remarks place these measurements against (2). First,  $\gamma_{\text{loc}}$  stays well above the lower bound  $1/(4+\Gamma)$  everywhere (the smallest measured  $\gamma_{\text{loc}} \approx 0.63$  versus a bound  $\leq 0.25$ ), as it must. Second, in this construction the smoothness satisfies  $\beta = 2\kappa/(\kappa + 1) < 2$ , so the  $\beta - 1$  branch of  $\Gamma$  is the active one for  $N_\theta \geq 2$ ; the experiments therefore do not drive the  $N_\theta(3 + \frac{4}{\sqrt{\pi}}(1 + \log \kappa))$  branch through  $\beta$ . What they test is the complementary and more basic question—whether the *realized* rate carries any  $N_\theta$  dependence at all—and the answer over the tested range is that it does not.

**The full-operator norm is a noisy diagnostic, not the rate.** A natural temptation is to read dimension dependence off the largest eigenvalue  $\hat{\Lambda}$  of the symmetric-matrix operator  $H_{\text{sym}}$ . This is misleading.  $H_{\text{sym}}$  acts on the  $N_\theta(N_\theta + 1)/2$ -dimensional space of symmetric matrices, so its top eigenvalue is biased upward by finite-sample spectral noise that grows with that dimension. Figure 4 makes this explicit for the separable family, where the Gauss–Hermite benchmark  $\Lambda_{\text{exact}}$  is exact and dimension-free: the Monte Carlo estimate  $\hat{\Lambda}_{\text{full sym}}$  rises with  $N_\theta$  while  $\Lambda_{\text{exact}}$  is flat. The gap is estimator noise, not a property of the flow, which is why the dimension-independence conclusion is drawn from the well-conditioned  $\gamma_{\text{loc}}$  (a smallest-eigenvalue problem on the metric-correct operator) and not from  $\hat{\Lambda}$ . In the revised diagnostics,  $\tau_H^2$  is tracked separately: unlike  $\hat{\Lambda}$ , it measures only the first-Hermite mean-covariance coupling, and the row-wise check  $\gamma_{\text{loc}} \geq 1/(\tau_H^2 + 3)$  is an algebraic consequence of the Bochner-Schur identity on the same operator.

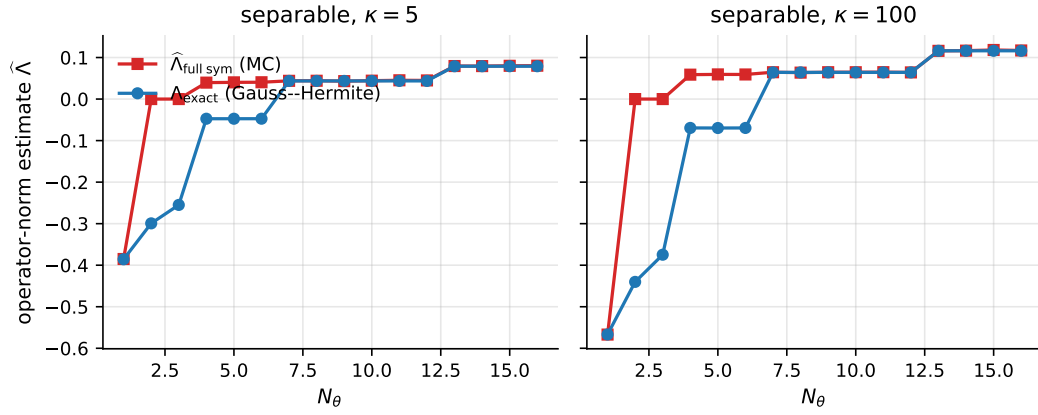


**Figure 3:** Local rate  $\gamma_{loc}$  versus  $N_\theta$ , one panel per conditioning  $\kappa$ , all families. The flatness in  $N_\theta$  holds at every  $\kappa$ ; the only systematic motion is the downward shift of the coupled families as  $\kappa$  increases.

## 4 Discussion

The measured local rate  $\gamma_{loc}$  is flat in  $N_\theta$  at every conditioning and moves only with  $\kappa$  (Figures 1–3). This is the behavior of the dimension-free form  $1/(4+D_\kappa)$  and not of the  $1/(4+n(\cdots))$  branch of the present bound (2). The numerical evidence therefore indicates that the genuine local convergence rate of Gaussian natural gradient flow is governed by the conditioning  $\kappa$ , and that the explicit  $N_\theta$  factor in the current proof is an artifact of the bounding argument—most plausibly the step that controls  $\lambda_{\max}(G - \mathbf{1}\mathbf{1}^\top)$  by  $N_\theta(3 + \frac{4}{\sqrt{\pi}}(1 + \log \kappa))$  via a sum over all coordinates, rather than a property of  $L_\star$  itself.

**Limitations.** This is numerical evidence, not a theorem. The grid covers  $N_\theta \leq 16$  at a single sample size  $M = 2^{22}$ ; while the separable diagonal and Gauss–Hermite benchmarks confirm that  $\gamma_{loc}$  is well-converged at this  $M$  (unlike  $\hat{\Lambda}$ ), the experiments do not establish behavior for  $N_\theta \gg 16$ . The conclusion is that the data are inconsistent with a genuine  $1/N_\theta$ -type decay over the tested range and consistent with a  $\kappa$ -only rate; turning this into a dimension-free guarantee requires a revised proof. The full-operator norm  $\hat{\Lambda}$  is reported only as a diagnostic and is explicitly not used as evidence about the rate.



**Figure 4:** Separable family: the Monte Carlo full-operator norm  $\hat{\Lambda}_{\text{full sym}}$  (red) inflates with  $N_\theta$ , whereas the exact Gauss-Hermite benchmark  $\Lambda_{\text{exact}}$  (blue), which is dimension-free by construction, is flat. The inflation is finite-sample spectral noise on the  $O(N_\theta^2)$ -dimensional operator and must not be read as a dimensional property of the rate.